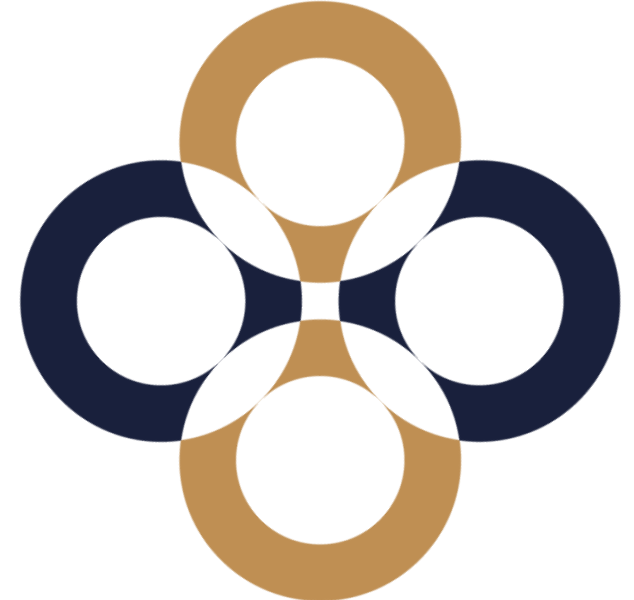


Capital Structure III. – Payout policy

Applied Corporate Finance – 6

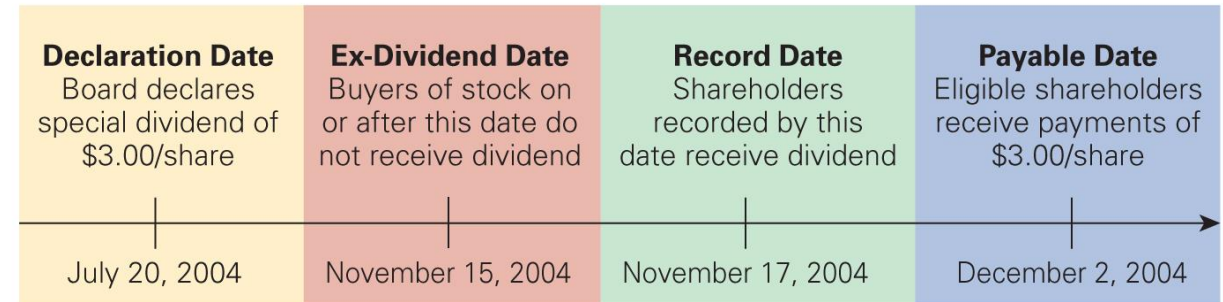
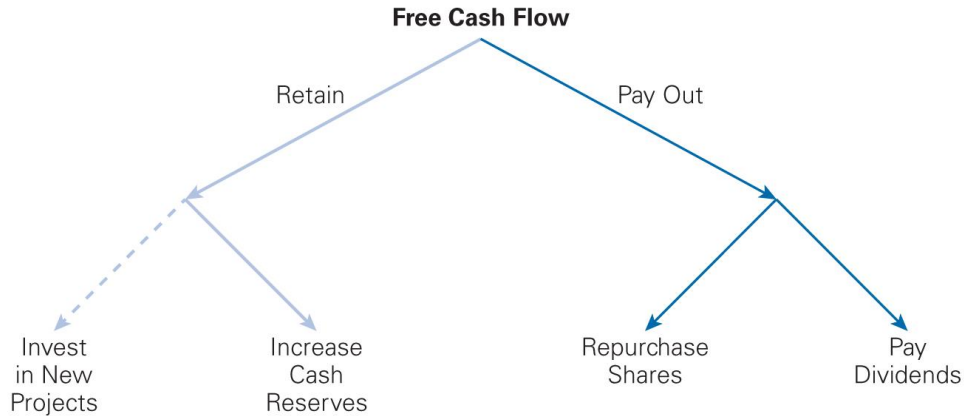
*Kata Varádi, Leyla Yusifzada, Noémie Cabau
Corvinus University of Budapest, Institute of Finance*

*Based on:
Berk – DeMarzo (2017): Corporate Finance. chapter 17*



Payout policy

Payout policy: the way a firm chooses between the alternative ways to distribute free cash flows to equity holders.



Declaration Date: The date on which the board of directors authorizes the payment of a dividend.

Record Date: When a firm pays a dividend, only shareholders on record on this date receive the dividend.

Ex-dividend Date: A date, two days prior to a dividend's record date, on or after which anyone buying the stock will not be eligible for the dividend

Payable Date (Distribution Date): A date, generally within a month after the record date, on which a firm mails dividend checks to its registered stockholders

Dividends

- Special Dividend:

A one-time dividend payment a firm makes, which is usually much larger than a regular dividend.

- Stock Split (Stock Dividend):

When a company issues a dividend in shares of stock rather than cash to its shareholders.

- Return of Capital:

When a firm, instead of paying dividends out of current earnings (or accumulated retained earnings), pays dividends from other sources, such as the liquidation of assets.

- Liquidating Dividend:

A return of capital to shareholders from a business operation that is being terminated.

Share repurchases

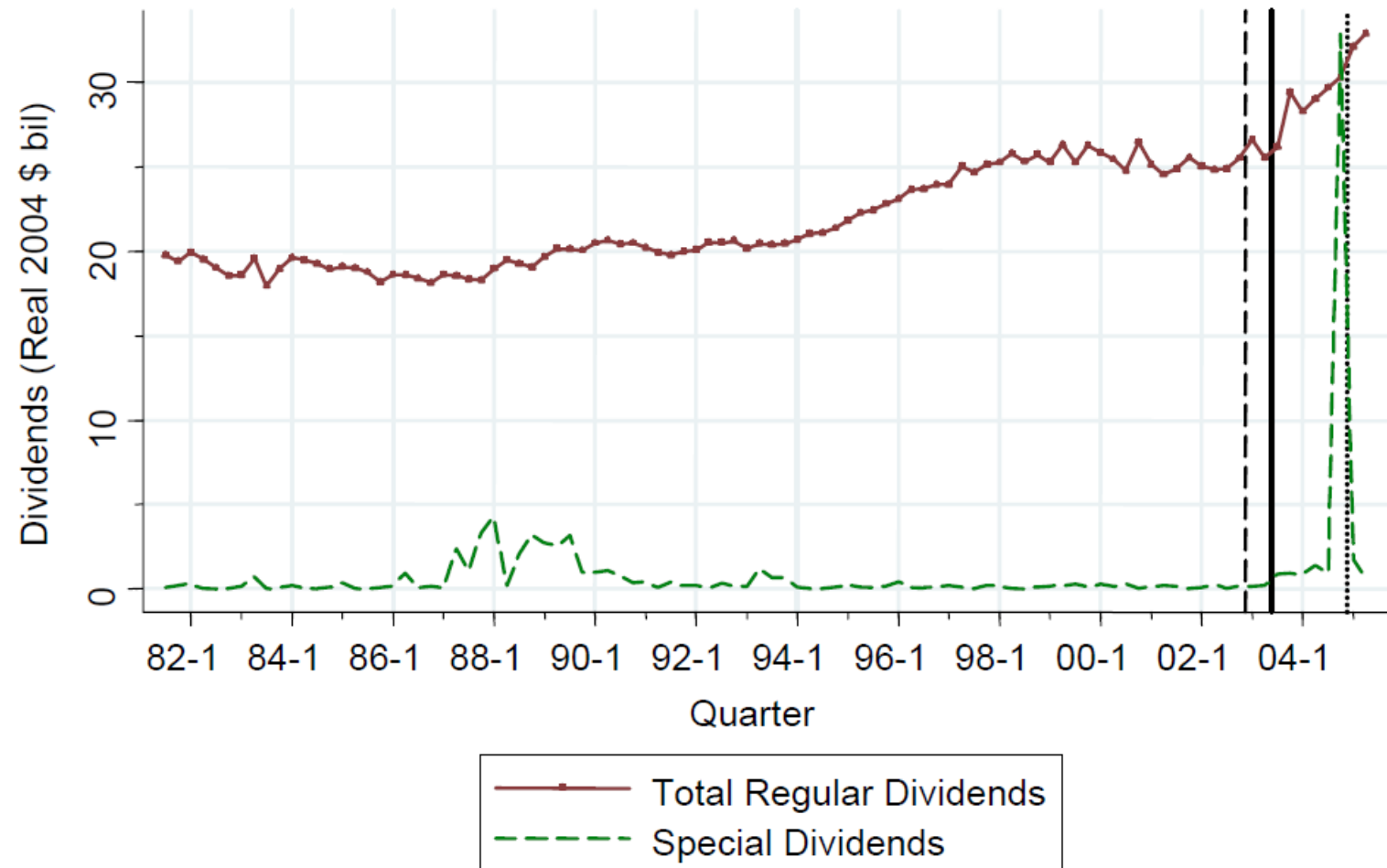
An alternative way to pay cash to investors is through a share repurchase or buyback.

- Open Market Repurchase: a firm repurchases shares by buying shares in the open market (95% of the cases)
- Tender Offer: A public announcement of an offer to all existing security holders to buy back a specified amount of outstanding securities at a pre-specified price (typically set at a 10% to 20% premium to the current market price) over a pre-specified period of time (usually about 20 days). If shareholders do not tender enough shares, the firm may cancel the offer, and no buyback occurs.
- Dutch Auction: the firm lists different prices at which it is prepared to buy shares, and shareholders in turn indicate how many shares they are willing to sell at each price. The firm then pays the lowest price at which it can buy back its desired number of shares.
- Targeted Repurchase: a firm purchases shares directly from a specific shareholder.
- Greenmail: a firm avoids a threat of takeover and removal of its management by a major shareholder by buying out the shareholder, often at a large premium over the current market price.

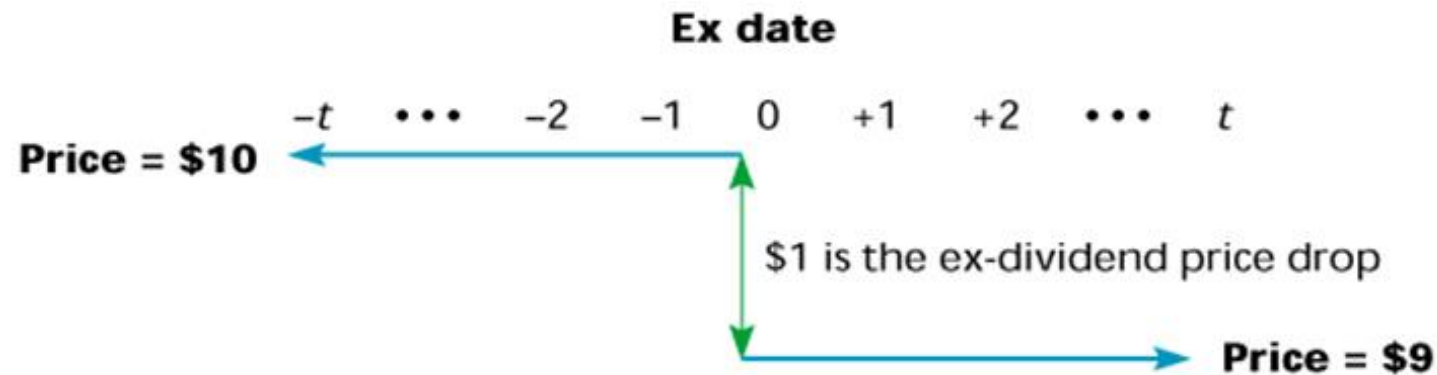
Example: Microsoft

- July 2004: FCF \$60bil
- Increased annual dividend to 0.32 ps
- Repurchase about \$30bil of company's stock in 4 years
- Make a special dividend payment of 3 dollars per share
- NOSH= 10.7 bil shares
- Special Dividend =32.6 bil = Largest corporate cash disbursement in history
- Total dividends paid by all S&P500 companies totaled 213.6 bil
- Accounted for more than 3% of all monthly personal income in the US

Example: Microsoft



Ex Dividend Price Drop



At the Ex Date the stock is trading without the dividend and in equilibrium its stock price should fall by the amount of the dividend

Shareholder's wealth is not modified by the dividends they receive. Dividends simply modify the shareholders' wealth composition

Measures of Dividend Policy

Dividend yield

Relates the dividend paid to the price of the stock

Annual dividends per share/price per share

A measure of the component of total return to shareholders that comes from dividends

Expected return on stock = Dividend yield + Price appreciation

The rate of growth of dividends per share is an important indicator of dividend policy (and earnings growth).

Dividend payout ratio:

Measures the percentage of earnings that the firm pays in dividends

Dividends/Earnings

Optimal Dividend Policy

Does payout policy affect shareholder value?

Early Models of Dividend Policy

A “not so good” reason why the higher the dividend, the higher the value of the firm:

The 'bird in the hand' argument

Individuals desire certain current income, rather than uncertain delayed returns, and thus bid up the price of stock paying high dividends

Fallacy

An increase in dividend payment to existing shareholders results in an equivalent future drop of the share of V of these same shareholders (at the ex-dividend date).

Modigliani–Miller and Dividend Policy Irrelevance

- There is a trade-off between current and future dividends.

If a firm pays a higher current dividend, future dividends will be lower.

If a firm pays a lower current dividend, future dividends will be higher.

- MM Dividend Irrelevance

In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and does not affect the initial share price.

Modigliani–Miller and Dividend Policy Irrelevance

- Thus, if there are no taxes, or if dividends and capital gains are taxed at the same rate, investors should be *indifferent to receiving their returns in dividends or price appreciation*.
- For this argument to work, in addition to assuming that there is no tax advantage or disadvantage associated with dividends, *we also have to assume the following*
 - No transaction costs
 - The investment decisions of the firm are unaffected by its dividend decisions
 - Managers do not waste cash

Comparison

In a perfect capital market...

- ...when a dividend is paid, the share price drops by the amount of the dividend when the stock begins to trade ex-dividend.
- ...an open market share repurchase has no effect on the stock price, and the stock price is the same as the cum-dividend price if a dividend were paid instead.



Investor Preferences: investors are indifferent between the firm distributing funds via dividends or share repurchases. By reinvesting dividends or selling shares, they can replicate either payout method on their own. → Home-made dividend

Tax disadvantage of dividends

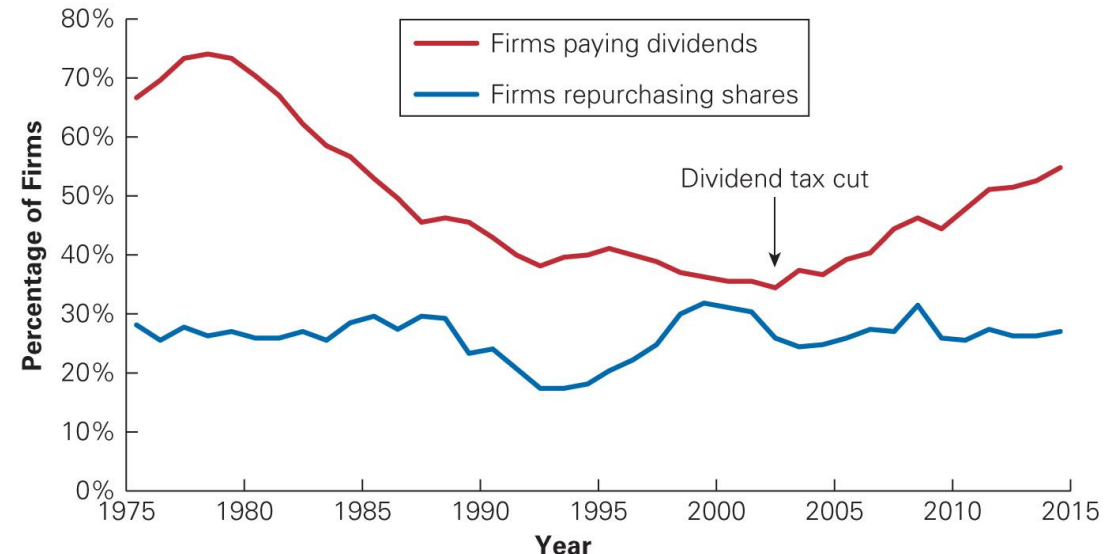
Taxes on Dividends and Capital Gains

- Shareholders must pay taxes:
 - on the dividends they receive,
 - on capital gains taxes when they sell their shares.

Typically:
taxes on dividend > taxes on capital gain.
Moreover, long-term investors can defer the capital gains tax forever by not selling.



- The higher tax rate on dividends makes it undesirable for a firm to raise funds to pay a dividend.
- **Dividend Puzzle:** When firms continue to issue dividends despite their tax disadvantage.



Dividends and Taxation

- Consider buying a stock just before it goes ex-dividend and selling the stock just after.

The equilibrium condition must be

$$(P_{cum} - P_{ex})(1 - \tau_g) = Div(1 - \tau_d)$$

$$P_{cum} - P_{ex} = Div \times \left(\frac{1 - \tau_d}{1 - \tau_g} \right) = Div \times \left(1 - \frac{\tau_d - \tau_g}{1 - \tau_g} \right) = Div \times (1 - \tau_d^*)$$

Where P_{cum} is the cum-dividend price, P_{ex} is the ex-dividend price, τ_g is the capital gains rate tax, and τ_d is the dividend tax rate.

Dividends and Taxation

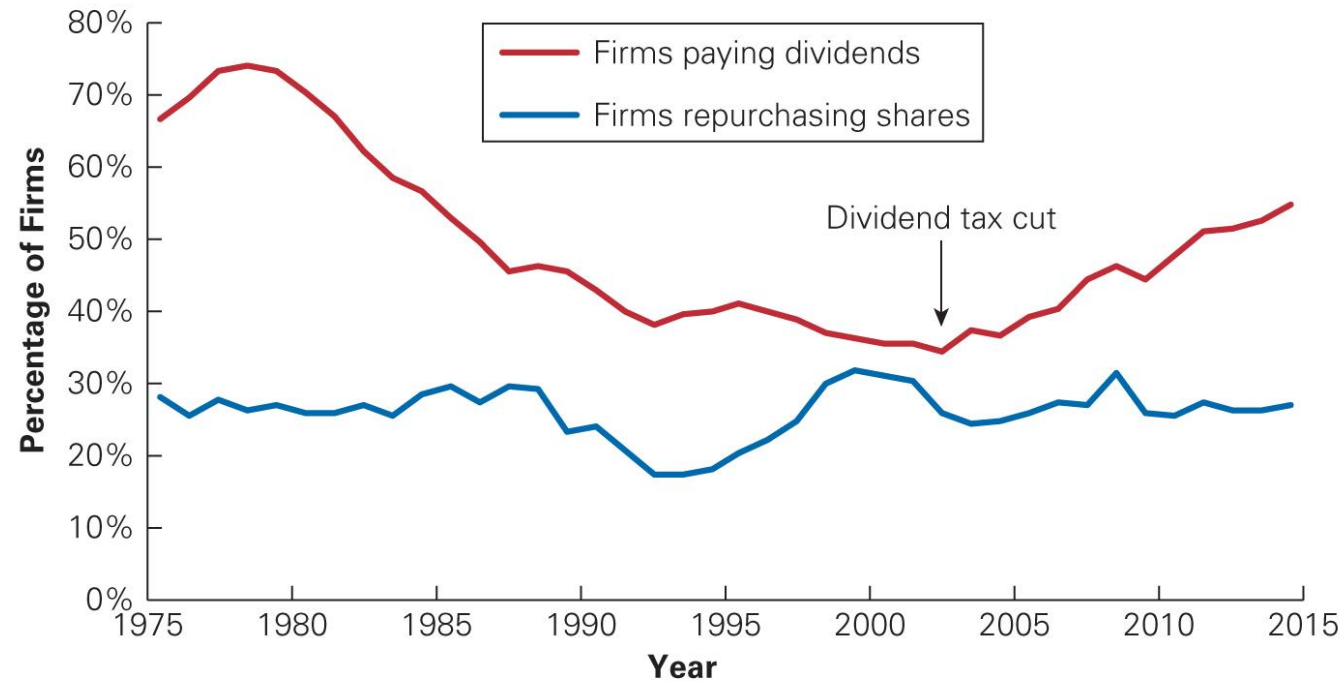
- When *the tax rate on dividends* > *the tax rate on capital gains*, shareholders prefer share repurchases
- When *the tax rate on dividends* = *the tax rate on capital gains*, still $PV(\text{capital gains tax}) < PV(\text{dividends tax})$.
 - since they are deferrable
- **If taxes matter then firms that pay dividends can be of lower value.**
- **Then it would be optimal for firms not to pay dividends at all.**
Black (1976) --“Dividend Puzzle”

TABLE 17.2**Long-Term Capital Gains Versus Dividend Tax Rates in the United States, 1971–2012**

Year	Capital Gains	Dividends
1971–1978	35%	70%
1979–1981	28%	70%
1982–1986	20%	50%
1987	28%	39%
1988–1990	28%	28%
1991–1992	28%	31%
1993–1996	28%	40%
1997–2000	20%	40%
2001–2002	20%	39%
2003–2012	15%	15%
2013*–	20%	20%

*The tax rates shown are for financial assets held for more than one year. For assets held one year or less, capital gains are taxed at the ordinary income tax rate (currently 39.6% for the highest bracket); the same is true for dividends if the assets are held for less than 61 days. Because the capital gains tax is not paid until the asset is sold, for assets held for longer than one year the *effective* capital gains tax rate is equal to the present value of the rate shown, when discounted by the after-tax risk-free interest rate for the additional number of years the asset is held.

Trends in the Use of Dividends and Repurchases



Why Do Firms Pay Dividends?

- Clientele effect
- To convey favourable information
- To reduce the agency problem

Clientele Effect

- ***Different investors pay different tax rates***: while some individuals and institutions pay no taxes, other pay varying levels of taxes.
- Those who pay no taxes and have a preference for current income *prefer dividends to capital gains*.
Institutional Investors: pension funds, university endowments, charitable foundations
- Those who have high marginal tax rates *prefer future capital gains (or repurchases) to dividends*
- Investment horizon also plays a role
Younger Investors with longer horizons prefer maximum capital growth to dividends

Tax differences across investors

- The effective dividend tax rate differs across investors for a variety of reasons.

- Type of Investor or Investment Account
- Income Level
- Investment Horizon
- Tax Jurisdiction

As a result of their different tax rates, investors will have varying preferences regarding dividends.

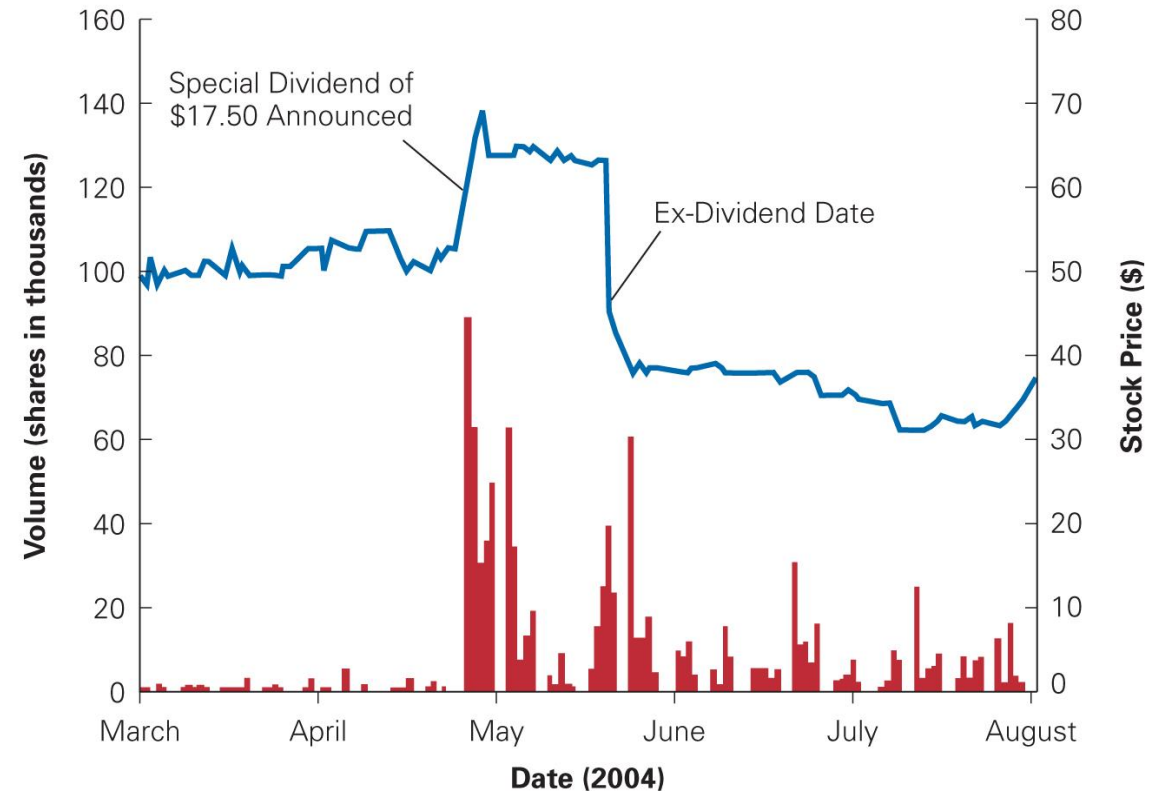
Investor Group	Dividend policy preference	Proportion of investors
Individual investors	Tax disadvantage for dividends generally prefer share repurchase(except for retirement accounts)	~52%
Institutions, pension funds	No tax preference prefer dividend policy that matches income needs	~47%
Corporations	Tax advantage for dividends	~1%

Source: Proportions based on **federal reserve flow of funds accounts**.

Clientele effect

Dividend-Capture Theory: absent transaction costs, investors can trade shares at the time of the dividend so that non-taxed investors receive the dividend.

An implication of this theory is that we should see large trading volume in a stock around the ex-dividend day, as high-tax investors sell and low-tax investors buy the stock in anticipation of the dividend, and then reverse those trades just after the ex-dividend date.



Signaling

- If there is information asymmetry dividends can be used as signals
- Firms can signal *positive information* through dividends.
- An increase in dividend payout improves market perceptions.
- The positive signalling value of dividends can be traded off against the tax loss (Bhattacharya, 1979)
- Their signal is credible since 'weaker' firms would have to finance the increased dividends with increased borrowing and thus reveal their nature.
- Dividend Cuts are very bad signals

Signaling with payout policy

Dividend Smoothing: Firm change dividends infrequently, and dividends are much less volatile than earnings.

Dividend Signaling Hypothesis: The idea that dividend changes reflect managers' views about a firm's future earning prospects → If firms smooth dividends, the firm's dividend choice will contain information regarding management's expectations of future earnings.

- When a firm increases its dividend, it sends a positive signal to investors that management expects to be able to afford the higher dividend for the foreseeable future.
- When a firm decreases its dividend, it may signal that management has given up hope that earnings will rebound in the near term and so need to reduce the dividend to save cash.

...payout increase might also signal a lack of investment opportunities. A firm might cut its dividend to exploit new positive-NPV investment opportunities. → In this case, the dividend decrease might lead to a positive, rather than negative, stock price reaction.

Signaling and share repurchase

Share repurchases are a credible signal that the shares are underpriced, because if they are overpriced a share repurchase is costly for current shareholders.

If investors believe that managers have better information regarding the firm's prospects and act on behalf of current shareholders, then investors will react favorably to share repurchase announcements.

Reduce agency problem

The Free Cash Flow Hypothesis (Jensen, 1986, AER)

- Excess cash can lead to agency problems if managers are empire builders.
 - Firms may be able to mitigate the agency problems by paying dividends
- Dividends are a way of getting money back to investors, preventing managers from wasting money on negative NPV projects

Other factors affecting dividend payout policy

- Investment opportunities
- Expected volatility of future earnings
- Financial flexibility
- Floatation costs
- Contractual and legal restrictions

Taxes and cash retention

Corporate taxes make it costly for a firm to retain excess cash → Cash is equivalent to negative leverage, so the **tax advantage of leverage** implies a **tax disadvantage to holding cash**.

Shareholder taxes: When a firm retains cash, it must pay corporate tax on the interest it earns. In addition, the investor will owe capital gains tax on the increased value of the firm. In essence, the interest on retained cash is taxed twice.

If the firm paid the cash to its shareholders instead, they could invest it and be taxed only once on the interest that they earn. The cost of retaining cash therefore depends on the combined effect of the corporate and capital gains taxes, compared to the single tax on interest income.

$$\tau_{retain}^* = \left[1 - \frac{(1 - \tau_c)(1 - \tau_g)}{(1 - \tau_i)} \right]$$

Issuance and distress costs

Generally, firms retain cash balances to cover potential future cash shortfalls, despite the tax disadvantage to retaining cash.

A firm might accumulate a large cash balance if there is a reasonable chance that future earnings will be insufficient to fund future positive-NPV investment opportunities.

The cost of holding cash to cover future potential cash needs should be compared to the reduction in transaction, agency, and adverse selection costs of raising new capital through new debt or equity issues.

Agency costs and retaining cash

When firms have excessive cash, managers may use the funds inefficiently by paying excessive executive perks, over-paying for acquisitions, etc. → payout can boost the stock price by reducing managers' ability and temptation to waste resources.

Firms should choose to retain to help with future growth opportunities and to avoid financial distress costs.

It is not surprising that high-tech and biotechnology firms tend to retain and accumulate large amounts of cash.

Ticket (2015)	Company	Cash & Marketable Securities(\$Billion)	Percentage of Market Capitalization
AAPL	Apple Inc.	215.7	37%
GE	General electronic	113.8	39%
MSFT	Microsoft corporation	102.3	23%
GOOGL	Alphabet (Google)	73.1	14%
CSCO	Cisco systems	59.1	43%
ORCL	Oracle corporation	52.3	34%
AMGN	Amgen, Inc.	31.4	26%
GM	General motors	20.3	38%

Stock dividends, splits

Stock dividend: With a stock dividend, a firm does not pay out any cash to shareholders.

- total market value of the firm's equity is unchanged.
- The only thing that is different is the number of shares outstanding.
- The stock price will therefore fall because the same total equity value is now divided over a larger number of shares.
- Stock dividends are not taxed.

Stock splits:

- The number of shares is proportionally increased, and the price per share is proportionally reduced so that there is no change in value.
- The typical motivation for a stock split is to keep the share price in a range thought to be attractive to small investors.
- On average, announcements of stock splits are associated with a 2% increase in the stock price.

Spin-offs

Spin-off: When a firm sells a subsidiary by selling shares in the subsidiary alone.

- Non-cash special dividends are commonly used to spin off assets or a subsidiary as a separate company.

Spin-Offs offer two advantages:

- It avoids the transaction costs associated with a subsidiary sale.
- The special dividend is not taxed as a cash distribution.



**Thank you
for your attention!**